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Three Distinct Step Count Trajectories Identified at 3-Months After Total Knee Replacement: An Exploratory Cohort Study

Total knee replacement (TKR) is an effective treatment for end-stage knee osteoarthritis (OA), leading to improved outcomes.⁷ Following TKR, physical activity is an important component of postoperative rehabilitation.³⁵ Physical activity comprises “any bodily movement produced by skeletal muscles that results in energy expenditure.”⁸ However, improvements in outcomes following TKR do not appear to correspond to increased levels of physical activity, with patients

following TKR completing less than age-matched controls^{15,30} and 42% not completing enough physical activity to maintain their health and fitness.¹⁵

Poor outcomes following TKR have been linked to a lack of physical activity, prior to the surgery, but also after.¹⁰ Not adhering to health care provider recommendations following TKR has been linked to postoperative complications such as increased pain and stiffness.³² Adherence to exercise and physical activity is complex and multidimensional,^{5,31} further complicated by the lack of a gold standard measure.^{3,11,27} There is currently limited information on adherence to physical activity following TKR, with the majority of studies poorly reporting or not reporting adherence.⁴

Previous studies have aimed to quantify the amount of physical activity completed following TKR by objectively measuring step count. The number of steps completed following TKR is varied^{9,15,26} and adherence to a step count intervention is likely to be influenced by numerous factors.^{12,33} Furthermore, previous studies have not categorized participants into different step count trajectory groups, which may explain the variation in outcomes among patients following TKR.^{2,16} The identification of step count trajectories may provide a greater understanding of how participants recover over time following TKR, and may also enable identification of participants at risk of low or declining step counts. The aim of this exploratory cohort study was to

■ **OBJECTIVE:** To determine if participants demonstrate distinct trajectories of step count following total knee replacement, and whether their baseline characteristics differ.

■ **DESIGN:** Longitudinal repeated measures study within a randomized-controlled trial.

■ **METHODS:** Participants were recruited from 3 hospitals in Sydney, Australia. Only participant data from the intervention group were used for this study (n = 51). Baseline data including demographics, pain (0-10 numerical rating scale), 13-item Patient Activation Measure scale, and modified Computer Self-efficacy Scale were collected. Participants received an activity tracker to monitor daily step counts. Latent class analyses were used to identify distinct trajectories of step count from baseline to 3-month follow-up. Baseline characteristics were compared using multinomial logistic regression.

■ **RESULTS:** 43 participants were included in the analysis. Participants were predominantly

female (n = 30, 70%) with an average age (SD) of 67.7 (7.5) years and body mass index (SD) of 30.3 (6.0). Three distinct step count trajectories were identified: a “high and rapidly increasing” group (n = 6, 14%), who began at 6251 ± 3508 steps and achieved 12794 ± 3173 steps by weeks 9 to 11; a “low and gradually increasing” (n = 24, 56%), (baseline: 2843 ± 1058 steps, 12 weeks: 6441 ± 1677 steps); and “high and gradually increasing” (n = 13, 30%), (baseline: 6299 ± 1777 steps, 12 weeks: 9614 ± 1436 steps). At baseline, body mass index and pain significantly differed between groups.

■ **CONCLUSION:** Three distinct step count trajectories were found in people with total knee replacement. Findings highlight the variability between patient groups and the need for support for those who may not be reaching prescribed step goals. *JOSPT Open* 2023;1(1):78-85. *Epub* 11 August 2023. doi:10.2519/josptopen.2023.0011

■ **KEY WORDS:** digital health, knee replacement, osteoarthritis, physical activity, step count

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determine if there were distinct step count trajectories in participants following TKR and whether their baseline characteristics differed.

METHODS

Design

This study was a longitudinal repeated-measures cohort study nested within the PATHway (participatory health through behavioral engagement and disruptive digital technology for postoperative rehabilitation) randomized controlled trial. PATHway is a multicenter randomized controlled 2-arm trial with 1:1 allocation ratio, parallel, assessor, and statistician-blinded superiority, comparing disruptive digital technologies integrated into a postsurgical pathway compared to usual care in persons undergoing TKR or lumbar decompression.³⁶ The PATHway trial includes a 6-month intervention period and a 6-month follow-up period, with the primary outcome measured at 3 months.¹¹ Only data from participants randomized to the TKR intervention group are used in the current analyses. Ethical approval for PATHway was granted from the University of Sydney Human Research Ethics Committee (protocol 2018/265). All participants provided written informed consent prior to randomization.

Participants

The inclusion criteria for participants with TKR were age ≥ 18 years, admitted for rehabilitation following TKR, familiarity with the use of the Internet and smart devices, ownership of a smartphone; and the ability and willingness to complete study procedures for 12 months. The exclusion criteria were comorbid health conditions that would prevent active participation in the physical activity program; inadequate English to provide fully informed consent, participate in health coaching sessions, or complete outcome measures; undergone

revision or bilateral TKR; taking part in another clinical trial or current use of any intervention component; and unwillingness to stop for the duration of the study.

Intervention

Participants in the intervention group received a 6-month digital technology package designed to optimize engagement and adherence to the TKR rehabilitation program. This program was provided in addition to the standard care that they were receiving (ie, from outpatient rehabilitation day program services). The technology package consisted of an exercise program application, a wearable fitness tracker (Fitbit Inspire, Fitbit, USA), and fortnightly health coaching sessions. Health coaching over 3 months was carried out by the study coordinator (V.D. – physiotherapist with experience in postoperative and rehabilitation settings) who also received formal training and qualifications as a health coach (Foundations of Health and Wellness Coaching – Levels 1 and 2, delivered by Wellness Coaching Australia consisting of 20 hours of foundation theory learning and practical application skills). Additional information about the intervention can be found in the published protocol.³⁶

Measures

Step Count Activity Step counts were exported from individual participant Fitbit accounts as CSV files at each fortnightly health coaching session. The date from the current day to previous 14 days was set as the date range. Step count activity was measured by calculating the average number of steps over the previous 14 days in the exported CSV file by using the average function in Excel. Fitbit devices have shown to provide reliable and validated measures of step count using 7-day step averages in community-dwelling older adults.²⁹ We decided to extend the date range to 14 days to capture variations in

activities over weekdays and 2 weekends, which may have arisen due to occupational, social, or family commitments. All step count data for each participant were checked prior to analysis. Missing data or outlier data points were omitted from the step count average, and step count was only averaged over the number of days with full data. As a guideline, participants were provided a target number of steps during each health-coaching session; participants were encouraged to complete at least 1000, 4500, and 7500 steps per day from weeks 1 to 5, 6 to 11, and >12 , respectively. The step goals were based on previous literature reviews documenting step count following TKR,^{9,25,34} and recent evidence suggesting that 7500 steps per day reduces all-cause mortality in older women.²⁴

Demographic Information Demographic information, including age, sex, and body mass index (BMI), was collected at baseline. BMI was collected by the hospital site staff from participant admission records.

Pain Baseline pain was self-reported by participants using a 0-10 numerical rating scale (NRS), where 0 indicates no pain and 10 indicates the worst possible pain.

Patient Activation The 13-item Patient Activation Measure (PAM-13) survey was used to assess patient activation—an individual's knowledge, skills, and confidence to manage their health condition.¹⁷ The results of the PAM survey provide 2 measures of activation: a raw score (ranging from 0 to 100, but usually between 30 and 90) and a level (from 1 to 4, where 1 = “disengaged and overwhelmed”; 2 = “becoming aware, but still struggling”; 3 = “taking action”; and 4 = “maintaining behaviors and pushing further”).¹⁷ The PAM has been validated as an acceptable measure of self-management capability in an OA population (Cronbach's $\alpha = 0.92$, person reliability = 0.98, item reliability = 0.87).¹³

Technology Self-efficacy Technology self-efficacy was measured using the modified Computer Self-efficacy Scale (mCSES), which consists of 10 questions assessing confidence using a new technology.²³ Participants are asked to rate their confidence for each question on a scale from 0 to 10 (where 0 = “not at all confident” and 10 = “completely confident”) to provide a total score ranging from 0 to 100, with higher scores representing higher technology self-efficacy. The mCSES has been validated in a clinical rehabilitation setting for older adults in Australia (standardized $\alpha = 0.94$).²³

Data Analysis

The baseline characteristics for the total and individual groups were summarized using means and standard deviations (SDs) for continuous variables. Categorical variables were summarized using proportions and counts. Latent class growth analyses were fitted to the step count data using the “lcmm” package in R. Latent class growth analyses were used to detect subgroups of different step count trajectories over time and can be used to identify distinct trajectories within a larger heterogeneous population.²¹ The dates from each fortnightly health coaching session were converted to an approximate time point and analyzed, where time point 1 through 6 correspond with ≤ 3 weeks, >3 weeks ≤ 5 , >5 weeks ≤ 7 , >7 weeks ≤ 9 , >9 weeks ≤ 11 , and >12 weeks, postrandomization, respectively. Although most health coaching sessions took place within 3-month postrandomization, a sixth time point of “ >12 weeks” was created to account for scheduling variations, where some participants completed their final health coaching session after 12 weeks.

Three different models ranging from 1 to 5 groups were fitted: a fixed-effects model, a random intercept model (mixture model with random intercept), and a ran-

dom coefficient model (mixture model with random intercept and slope). The final model was selected based on a combination of Bayesian information criterion (BIC), a measure of model performance based on model discrepancy and complexity, with lower values indicating a relatively parsimonious model, and the percentage of participants belonging to each group (minimum of 10%). Following selection of the optimal model, baseline characteristics between the groups were compared using multinomial logistic regression, with a significance level set to 5%.

RESULTS

Study Population

A total of 43 participants were included in the longitudinal analyses. Four participants withdrew from the study before 3 months ($n = 1$ due to personal circumstances, $n = 1$ due to change of mind, $n = 2$ were unable to be reached) and 4 participants withdrew from the intervention and did not have sufficient Fitbit data to be included in the analysis. All participants had at least 3 step count measures from time points 1 (≤ 3 weeks) to 3 (>5 weeks ≤ 7). At time points 4 (>7 weeks ≤ 9), 5 (>9 weeks ≤ 11), and 6 (>12 weeks), there were 42 (97.6%), 38 (88.4%), and 19 (44.2%) step count measures, respectively. Most of the sample was female ($n = 30$, 69.8%) with an average age (SD)

of 67.7 (7.5) years and BMI (SD) of 30.3 (6.0). The participants had an average (SD) baseline pain of 5.5 (2.7) on the 0-10 NRS, patient activation score of 66.3 (20.4), and technology self-efficacy score of 77.2 (21.2) (TABLE 1).

Identification of Step Count Trajectories

Fixed-effect and random intercept models were fitted with 1 to 5 groups to determine step count trajectories. The 3-group random intercept model, with a random intercept and slope, was selected as the final model based on the BIC value and a minimum of 10% of participants in each group. For the 3 groups identified, the probability of participants belonging to groups 1, 2, and 3 was 83%, 87%, and 77%, respectively, indicating that the majority of participants belonged clearly into 1 of 3 groups. The BIC values and group percentages for each model are reported in TABLE 2.

FIGURE 1 depicts the overall step count trajectories, as well as the trajectories identified by the latent class analysis, where the solid black line represents the local regression (loess) smoothed step count trajectories. The corresponding mean step counts and SDs for the overall cohort, as well as each individual group at each time point is displayed in TABLE 3. The optimal 3-group random intercept and slope model split participants into 3 groups based on their step counts: a first

TABLE 1

Baseline Characteristics of Participants in Each Trajectory Group and Univariate Analyses

Characteristic	All Participants	Group 1 (n = 6) High, Rapidly Increasing	Group 2 (n = 24) Low, Gradually Increasing	Group 3 (n = 13) High, Gradually Increasing	P Value for Differences Between Groups
Female, n (%)	30 (69.8)	5 (83.3)	19 (79.2)	6 (46.2)	0.091
Age, years (SD)	67.7 (7.5)	66.8 (8.9)	68.3 (6.8)	66.9 (8.7)	0.817
BMI (SD)	30.3 (6.0)	24.5 (4.1)	31.8 (5.8)	30.5 (5.8)	0.007 ^a
Baseline pain (SD)	5.5 (2.7)	3.0 (3.6)	6.1 (2.4)	5.4 (2.4)	0.040 ^a
PAM-13 score (SD)	66.3 (20.4)	72.4 (17.5)	65.9 (22.8)	64.2 (17.7)	0.703
mCSES score (SD)	72.2 (21.2)	60.7 (25.8)	70.1 (21.5)	81.3 (15.5)	0.091

Abbreviations: BMI, body mass index; mCSES, modified Computer Self-efficacy Scale; PAM-13, 13-item Patient Activation Measure; SD, standard deviation.
^aStatistically significant.

group whose step counts began high and rapidly increased, termed “high and rapidly increasing” (group 1, $n = 6$, 14% of the overall participants); a second group whose step counts started lower and slowly increased, termed “low and gradually

increasing” (group 2, $n = 24$, 56% of the overall participants); and a third group whose step counts began high, stayed fairly consistent, and then gradually increased, termed “high and gradually increasing” (group 3, $n = 13$, 30% of the

overall participants). Interestingly, the participants in groups 1 and 3 both began around 6300 steps at time point 1, and the participants in group 2 finished around this number of steps at time point 6.

Baseline Characteristics of Participants in the 3 Step Count Trajectory Groups

Baseline characteristics of participants in each trajectory group are presented in **TABLE 1**. There were 2 baseline characteristics that distinguished the 3 groups: BMI and baseline pain scores (**TABLE 1**). An increase of 1 unit in BMI was associated with 28% lesser odds of being in group 1 (“high and rapidly increasing adherence”) versus group 3 (“high and gradually increasing adherence”) (odds ratio = 0.72, 95% CI: 0.532, 0.975). The 3 groups differed in their baseline pain levels; however, the pairwise comparison was not statistically significant. **FIGURE 2** displays the mean age, proportion of participants in each trajectory group classified with “obesity” (BMI ≥ 30), high patient activation scores (PAM-13 score $\geq 55.2^{14}$), severe baseline pain (NRS $\geq 7^6$),

TABLE 2

Goodness-of-Fit Statistics for the Trajectory Models

Model	Number of Groups	BIC	Group, %	Posterior Probabilities
Fixed-effect model	1	1122.6	100	1.0
	2	1025.5	63, 37	0.96, 0.88
	3	965.3	42, 51, 7	0.94, 1.0, 1.0
	4	944.5	30, 26, 7, 37	0.85, 1.0, 1.0, 0.94
	5	944.5	7, 30, 26, 19, 19	1.0, 0.92, 0.82, 1.0, 0.75
Random intercept model (mixture model with a random intercept)	1	905.9	100	1.0
	2	893.9	93, 7	1.0, 1.0
	3	899.6	7, 28, 65	1.0, 0.55, 0.82
	4	907.7	16, 63, 7, 14	0.86, 0.82, 1.0, 0.67
	5	917.5	7, 14, 7, 60, 12	1.0, 0.67, 1.0, 0.92, 0.40
Random coefficient model (mixture model with a random intercept and slope)	1	904.3	100	1.0
	2	902.4	12, 88	1.0, 1.0
	3 ^a	912.3	14, 56, 30	0.83, 0.88, 0.77
	4	911.0	7, 60, 21, 12	1.0, 0.89, 0.78, 0.80
	5	925.8	12, 12, 23, 7, 47	0.80, 0.0, 0.7, 1.0, 0.1

Abbreviation: BIC, Bayesian information criterion (where the lowest value indicates the best fitting model).

^aFinal model selected.

FIGURE 1

Overall trajectory of step count over time and the 3 groups identified using loess regression.

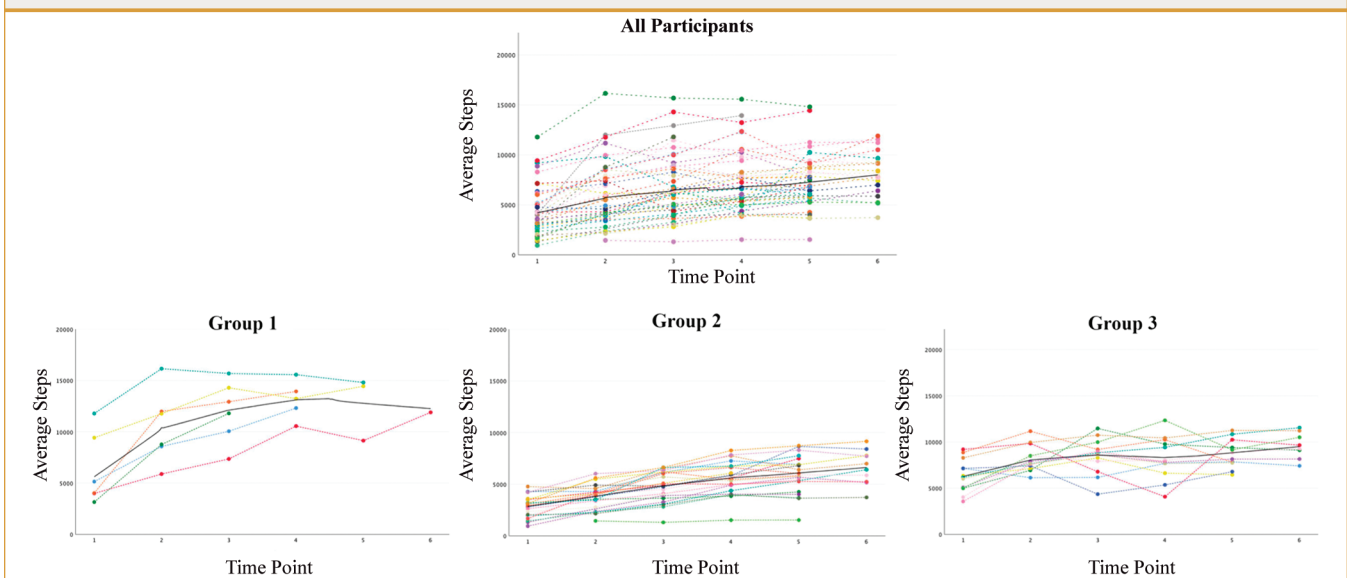


TABLE 3 Mean Step Counts at Each Time Point								
Time Point	All Participants		Group 1		Group 2		Group 3	
	N	Steps (SD)	N	Steps (SD)	N	Steps (SD)	N	Steps (SD)
1 (≤3 weeks)	41	4438 (2468)	6	6251 (3508)	22	2843 (1058)	13	6299 (1777)
2 (>3 weeks ≤ 5)	41	6178 (3189)	6	10520 (3572)	22	3810 (1208)	13	8182 (1402)
3 (>5 weeks ≤ 7)	43	6896 (3215)	6	12018 (3004)	24	4769 (1447)	13	8461 (1890)
4 (>7 weeks ≤ 9)	42	7294 (3046)	5	13120 (1864)	24	5560 (1529)	13	8256 (2213)
5 (>9 weeks ≤ 11)	38	7348 (2704)	3	12794 (3173)	23	5925 (1743)	12	8715 (1527)
6 (>12 weeks)	19	8169 (2281)	1	11893	10	6441 (1677)	8	9614 (1436)
Abbreviation: SD, standard deviation.								

and possessing average levels of technology self-efficacy (mCSES score $\geq 57^{23}$).

DISCUSSION

In our exploratory study, we found 3 distinct step count trajectories in participants following TKR over a time period of 12 weeks: a first group whose step count increased rapidly over the first 2 time points and remained high compared to the other groups (“high and rapidly increasing”); a second group that started lower than both groups and incrementally increased over each time point (“low and gradually increasing”), and a third group whose step counts began high and gradually increased at each time point (“high and gradually increasing”). The posterior probability values for the 3 step count adherence groups identified were high, indicating that the participants were appropriately classified into 1 of 3 groups. The baseline characteristics that distinguished the 3 groups were BMI and baseline pain scores. A 1-unit increase in BMI was associated with 28% lesser odds of being in the “high and rapidly increasing” versus the “high and gradually increasing” adherence group. The baseline pain scores of the 3 trajectory groups identified differed

($P = .040$); however, pairwise comparison was not statistically significant.

To our knowledge, this is the first study to examine step count trajectories in a post-TKR population. Previous studies have investigated step count trajectories in large samples of healthy adults.^{18,20,22} Our results are similar to a previous study of physical activity in women, where they found that those with higher body fat were more likely to be in the “low active” versus “somewhat active” and “active” groups.²² Trajectories of exercise adherence have also been investigated in an OA population, whereby participants with “rapidly declining adherence” had lower baseline pain and higher function at baseline compared to the “gradually declining adherence” group,²⁸ which contrasts with our findings. These differences may be due to different factors affecting adherence for physical activity compared to a prescribed exercise program, as well as the differences in patient characteristics (post-TKR versus chronic OA patients) and follow-up periods. Future studies should explore the presence of step count trajectories in a larger sample with different baseline characteristics, over a longer follow-up period.

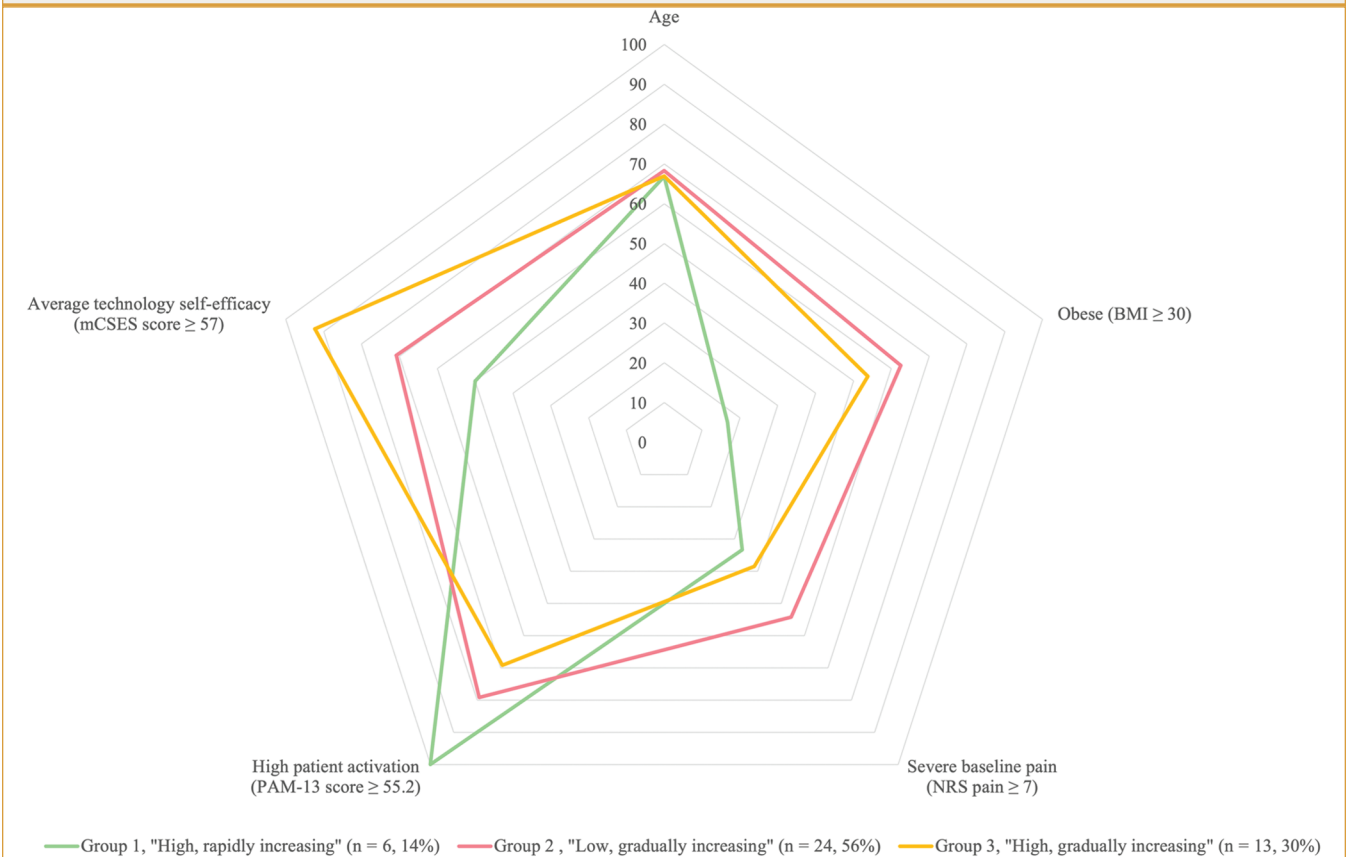
One strength of this study is that the step count data were objectively collected

using Fitbit devices. Fitbit devices are a commercial wearable tracker that has been increasingly used in research. The step counts generated from the Fitbit have been demonstrated to be comparable to those of research-grade accelerometers such as the SenseWear arm band (ICC = 0.79).¹⁹ Another strength is that instead of dichotomizing participants into “adherent” and “nonadherent” groups, we analyzed the raw step count data as a continuous outcome. Although systematic reviews investigating physical activity and exercise have found that the most used cutoff values defining “satisfactory” adherence are $\geq 80\%$ of completion,^{3,11} the interpretation of adherence is varied and subject to individual interpretation.

Limitations of this study include the small sample size and exploratory nature of latent class analyses. Of the 51 participants enrolled into the intervention group, only 43 participants were analyzed (84.3%). Undertaking the analysis in a greater sample size would help to confirm our findings. Although the selection of the number of latent classes, model fit index interpretation, and problem of convergence have been debated,²¹ latent class analysis is a widely accepted method and

FIGURE 2

Spidergram plot of identified trajectories showing age and proportion of each variable in each step count trajectory. Abbreviations: BMI, body mass index; mCSES, modified Computer Self-efficacy Scale; NRS, numerical rating scale; PAM-13, 13-item Patient Activation Measure.



is commonly used to explore trajectories of exercise and physical activity.^{22,28} Latent class analysis allows for differing levels of adherence over time, rather than assuming that all participants follow a common trajectory.²¹ Another limitation of the study is that it was nested within a larger RCT, which included health coaching as part of the intervention. The results of the study may not be generalizable to the wider population. The step counts and trajectories observed in this population may be due to participants' response to the health coaching, Fitbit device, and the feedback and motivation provided. The impact of adherence on step count trajectories should be investigated in a larger cohort, which was not possible in this small subgroup

analysis. Step counts were not measured prior to surgery; this additional information may have been useful to provide insights on physical activity levels prior to surgery and to assess recovery. Finally, although we encouraged participants to wear their Fitbits for 24 hours per day (only removing for daily bathing or weekly charging), we were unable to capture data on total Fitbit wear time. This may have resulted in step count discrepancies, with some activity uncaptured.

We provide preliminary evidence for researchers and physiotherapists that not all patients following TKR will progress at the same rate. Different subgroups of TKR patients such as those with higher BMI and baseline pain may progress slower in

terms of step count activity and may benefit from additional support. Additional support may come in the form of various behavior change techniques such as time management, social support, or monitoring to help promote physical activity.¹ Further work is warranted to explore whether similar trajectories in different populations exist and the additional effects of other baseline characteristics other than those examined in this study.

CONCLUSION

This study demonstrates marked variation in step counts following TKR. Recommendations for adherence to physical activity following surgery needs to be mindful of this variation and

differences in patient characteristics such as BMI and pain that might influence postoperative activity recovery. ■

KEY POINTS

FINDINGS: Our study identified 3 distinct step count trajectories in a cohort of participants who recently underwent total knee replacement, including a “high and rapidly increasing,” “low and gradually increasing,” and “high and gradually increasing” groups. Body mass index and baseline pain differed between the 3 groups.

IMPLICATIONS: Rehabilitation clinicians implementing a physical activity or step count intervention following total knee replacement should be mindful of individual variations in step count activity. Additional support may be provided to those who are not meeting their step goals.

CAUTION: We evaluated step count trajectories in a small sample of participants over 3 months. Further work is warranted to explore whether these trajectories exist in larger samples.

STUDY DETAILS

AUTHOR CONTRIBUTIONS: VD, PJAN, DJH conceptualized the study. VD wrote the first draft of the manuscript. VD, MLF, PJAN, XW and DJH acquired project funding. VD, SD and SRR contributed to project administration. SD, SRR and VV contributed to data curation. VD and VV conducted the formal analyses. All authors revised the manuscript critically for important intellectual content and read and approved the final version of the manuscript.

DATA SHARING: The deidentified data set is available upon request, through email to the corresponding author (vicky.duong@sydney.edu.au).

PATIENT AND PUBLIC INVOLVEMENT: Patient representatives and clinicians were consulted regarding the study design and procedures of the main trial.

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